SARA ALY

Los Angeles, CA • sarakhaled.kaz@gmail.com • (424) 460-9975
linkedin.com/in/sara-k-aly • sara-kaz.github.io

SUMMARY

Computer Engineering M.S. graduate (GPA 4.0, ARCS Fellow) with expertise in embedded AI, machine learning systems, and hardware–software co-design. Author of peer-reviewed publications in edge ML and distributed learning. Recipient of CSUN's university-wide Distinguished Thesis Award (1 of 4 selected from 18 theses across 9 colleges) and Outstanding Graduate Student Award (ECE). Seeking roles in embedded AI, ML engineering, or software engineering.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, C#, MATLAB, Verilog, SystemVerilog, VHDL

ML & AI: PyTorch, TensorFlow, Scikit-learn, Deep Learning, TinyML, Edge AI, Model Optimization, Distributed Machine Learning, Feature Engineering

Embedded & Hardware: ESP32, Jetson, Raspberry Pi, Seeed Studio XIAO nRF52840 (Sense), ARM Microcontrollers, FPGA, SoC Design, Real-Time Systems, I2C, SPI, UART, PlatformIO, Arduino

Software & Cloud: React, Django, REST APIs, PostgreSQL, MySQL, AWS RDS, Kubernetes, Power BI

Tools: Git, Bash, Azure DevOps, JIRA, Confluence, Vivado, Multisim, PSpice

EXPERIENCE

Graduate Research Assistant · *California State University, Northridge* — Sep 2024 – Present

- Researching hybrid CNN-Transformer architectures for image restoration tasks (denoising, deblurring, dehazing); two journal papers in progress
- Implemented and benchmarked Vision Transformer (ViT) models within Kubernetes clusters for scalable ML experimentation

Research Assistant — ARCS Research Program · *California State University, Northridge* — Sep 2024 – Present

- Designed fault-tolerant backup control architecture for autonomous drone navigation inspired by the NASA Mars Helicopter
- Modeled failure scenarios and hardware redundancy mechanisms to improve robustness under embedded hardware faults

Research Assistant — MOSAIC Lab · *California State University, Northridge* — Aug 2024 – Jan 2026

- Implemented distributed dual coordinate ascent algorithms in a ring-network topology for multi-node machine learning optimization
- Developed mathematical convergence models for distributed training under data and communication noise; results accepted to Asilomar 2025

Data Analyst Intern · *ConXtech* — Sep 2024 – Feb 2025

- Designed Power BI dashboards analyzing historical project cost estimates, reducing manual reporting overhead
- Built automated data pipelines using Excel, Access, and DAX formulas to enable real-time project analytics
- Created interactive visualizations used by engineering and project management teams for faster decision-making

AI/ML Engineer Intern · *PM Accelerator* — Sep 2024 – Dec 2024

- Built scalable backend database architecture using AWS RDS, PostgreSQL, and Django supporting multi-user data exchange workflows
- Developed dynamic scheduling and mentorship tracking platform frontend using React

Teaching Assistant · *California State University, Northridge* — Aug 2023 – May 2024

- Assisted with grading assignments and providing constructive feedback to students to support academic growth
- Monitored student progress and reported observations to the lead professor to inform instructional decisions
- Provided one-on-one support to students needing additional help with course concepts and assignments

Data Analyst · *Unipal* — Jan 2021 – Jan 2022

- Developed structured datasets covering universities, retail, and student services to support product launch
- Translated survey responses into statistical insights and visual dashboards for product and business teams

PROJECTS

Hardware-Accelerated Deep Learning for Multimodal Biomedical Monitoring with On-Device Adaptive Learning *— Thesis · Defended May 2026 · Patent Pending*

- Designed a multi-task CNN for simultaneous activity, stress, and arrhythmia classification running entirely on an ESP32-S3 microcontroller with no cloud dependency
- Developed an on-device adaptive training system in C++ ($\sim$1,200 lines) using head-only backpropagation (4,984 trainable vs. 107,568 frozen parameters), achieving a 715$\times$ reduction in activation storage over full backprop
- Persists personalized weights to NVS flash across device reboots within 300 KB heap and 8 MB PSRAM — *patent pending*

Smart Prosthetic Arm System *— CSUN · Aug 2023 – Jun 2024*

- Integrated a Bluetooth-enabled foot controller to a biomechanical prosthetic arm prototype, enabling wireless foot-based actuation via embedded C++ firmware
- Developed wireless communication stack between foot controller microcontrollers and arm hardware; maximum transmission delay under 165 ms
- Published in *Technologies*, vol. 13, no. 3, MDPI, Mar. 2025

TERM: Trustworthy Embodied Robotics Modeling with Inference Reasoning *— VLA Research · 2025 – Present · Under Review AAAI 2027*

- Designed 5-stream VLA architecture (VERA) fusing LLaMA and ViLT encoders across vision, instruction, and action-history inputs (∼96M parameters)
- Evaluated on Language-Table and CALVIN benchmarks; paper submitted to AAAI 2027

LLM & API Projects *— 2024 – Present*

- **AHMC Clinical HPI Generator:** LLM pipeline converting ER notes and H&P records into structured clinical summaries with an interactive clarification loop
- **GitHub Repository Analyzer:** Claude API-powered semantic code search and Q&A over arbitrary repositories
- **Pretty Good AI Voice Bot:** Real-time phone voice agent using OpenAI Realtime API and Twilio for end-to-end conversational voice handling

Parkinson's Disease Diagnosis using Machine Learning *— AUBH · Sep 2021 – May 2022*

- Developed and evaluated ML classifiers (logistic regression, SVM, weighted KNN) in MATLAB for early Parkinson's detection from patient voice recordings
- Quadratic SVM achieved best diagnostic accuracy; published in *Proc. AIVR 2022*, Springer Nature, 2023

EDUCATION

M.S., Computer Engineering · *California State University, Northridge* May 2026

GPA: 4.00 • ARCS Fellowship • Teaching Assistantship (Tuition Waiver) • Dean's List

Distinguished Thesis Award — University-wide, 1 of 4 selected from 18 theses across 9 colleges • **Outstanding Graduate Student Award**, ECE Dept. • Distinguished Graduate Student, Honors Convocation • 1st Place, CSUNposium Research Competition • Outstanding Student in Computer Engineering, Anagnost College of Engineering • Nominated as Commencement Speaker, College of Engineering & CS

Coursework: Deep Learning, System-on-Chip Design, ASIC Design, FPGA Systems

B.S., Computer Engineering · *California State University, Northridge* 2024

GPA: 3.94 • Summa Cum Laude • Dean's List • Owens Engineering Scholarship

Coursework: Data Structures & Algorithms, Machine Learning, Embedded Systems, Software Engineering

PUBLICATIONS

- **S. Aly** et al., "TERM: Trustworthy Embodied Robotics Modeling with Inference Reasoning," submitted to *AAAI Conf. on Artificial Intelligence*, 2027. *Under review.*
- **S. Aly** and S. Mirzaei, "Hardware-Aware Edge Deep Learning for Multimodal Biomedical Monitoring on Resource-Constrained SoCs," *IEEE Int'l Conf. on Healthcare Informatics (ICHI)*, 2026. **Best Paper Award**
- M. Cho, **S. Ali**, et al., "Distributed Dual Coordinate Ascent in a Ring Network for Distributed Machine Learning," in *Proc. Asilomar Conf. on Signals, Systems & Computers*, 2026.
- P.L. Bishay, G. Funes Alfaro, I. Sherrill, I. Reoyo, E. McMahon, C. Carter, C. Valdez, N.M. Riyaz, **S. Ali**, A. Lima, A. Nieto, J. Tirone, "The Foot Can Do It: Controlling the 'Persistence' Prosthetic Arm Using the 'Infinity-2' Foot Controller," *Technologies*, vol. 13, no. 3, p. 98, Mar. 2025. doi:10.3390/technologies13030098
- **S.K. Abdelhakeem**, Z.M. Mustafa, H. Kadhem, "Diagnosing Parkinson's Disease Based on Voice Recordings: Comparative Study Using Machine Learning Techniques," in *Proc. 6th Int'l Conf. on AI & Virtual Reality (AIVR 2022)*, Springer Nature, pp. 49–60, 2023.

LANGUAGES

English (Fluent) • Arabic (Native) • Korean (Intermediate)